Module 1 Reflection | Randi Weinstein | 6/20/2026

Notes from Reading 1

Guided reading questions (annotate answers as you go):

Q1: How do Bommasani et al. define a 'foundation model'? What two properties (training approach + adaptation range) are central to the definition?

Answer: Training is what is fed into model, adaptation is how it is fine-tuned to focus on a particular goal.

Q2: What does the paper mean by 'emergence,' and why does this property make foundation models difficult to predict even for their creators?

Answer: Emergence is the unintentional predictive part

Q3: The authors identify 'homogenization' as a double-edged property. What are the opportunity and the risk they associate with it?

Answer: Homogenization is creating a model that can be used for a wide range of tasks. The risk is that there may be inherent bias or misinformation that can cause it to be unreliable for all tasks.

Q4: Which example capability mentioned in Section 2 is most directly relevant to work in your professional domain? Why?

Answer: Reasoning and interaction - the why isn't as important as the reasoning behind how the model arrives at a conclusion; the point of all of this is to create output that can be useful to the human user

Key terms -- write your own definition after reading:

Term	Your definition (in your own words)
Foundational model	A model trained on broad data that can be adapted to a wide range of tasks
Fine-tuning	The adaptation part (specialization for a specific task, such as medicine or law)

In-context learning	A newer method of adapting a foundation model without requiring additional training (self-learning?)
Emergence	The behavior of a system that is unintentionally induced (unexpected outcomes) rather than intentionally constructed (expected outcomes)
Homogenization	Consolidation of methodologies for building machine learning systems across a wide range of applications A wide range of applications can be powered by a single generic learning algorithm Powerful, but potentially creates single points of failure or exposes flaws Aggressive homogenization is risky business (p 6)

Notes from Reading 2

Guided reading questions (annotate answers as you go):

- Q1: What benchmark domains show the largest AI performance gains in the most recent reporting year?

Answer: Answer: Math (International Math Olympiad)

- Q2: According to the Index, how has workplace adoption of generative AI changed among knowledge workers year-over-year?

Answer: Customer support and software developer jobs have been impacted the most (job loss, but also increased productivity)

- Q3: What does the Index report about AI-related educational offerings (courses, degrees, training programs)?

Formal education is lagging, but informal adoption is widespread

- Q4: What is one finding from the AI Index that you would characterize as a reason for caution rather than enthusiasm?

AI is transforming clinical care, but there is limited information on whether this is producing benefits to the general public in terms of real-life clinical scenarios

Key terms -- write your own definition after reading:

Term	Your definition (in your own words)
Benchmark	Test used to evaluate current status of a model
Frontier model	The most advanced models currently available for testing
Reasoning benchmark (e.g., MMLU. GPQA)	Massive Multitask Language Understanding; Graduate-Level Google-Proof Q&A Benchmark These are tasks intended to evaluate logic and reasoning rather than information recall
AI adoption rate	The speed at which the public is incorporating AI into their workflow
Compute scaling	The amount of GPU and energy to meet current demands

Notes from Task 1: Guided Reading with Active Annotation [60 min]

Step-by-step instructions:

1. Open both readings and your Annotation Table below (print or keep open in a second window).

2. Read Bommasani et al. Section 1 (pp. 1–12). Stop after every 2–3 paragraphs and complete one row.
3. Take a 5-minute break. Do not continue without it — spacing improves consolidation.
4. Read Bommasani et al. Section 2 overview (pp. 12–19). Add 3 more rows.
5. Read the AI Index Executive Summary + Chapters 4 & 6. Add 3 final rows.
6. Review your table. Circle the 2 rows that most directly connect to your professional work.
7. Flag any row where your 'Question' column remains unanswered — bring it to the synchronous session.

Annotation Table -- complete 8-10 rows across both readings

Source & page	Claim made by the authors	Evidence or support cited	Questions it raises for my work
Bommasani et al. Section 1, page 3-4	Early AI required algorithms, while modern AI (machine learning) is predictive.	Rather than specifying how to complete a task, modern AI learns (LLMs) how to complete the task. LLMs can do this for a wide range of tasks.	Perhaps some LLMs are better for specific tasks than others?
Bommasani et al. Section 1, page 4	Foundation models are successful due to “transfer of learning” and “scale”	Transfer of learning is relative to predictive ability. Scale results from better GPUs, transformer architecture, and more data.	Concern about copyright issues (training based on scientific publications without consent of authors).
Bommasani et al. Section 1, page 6	Homogenization comes with both pros and cons.	Homogenization allows generalization of model output, however it is also risky because	“All purpose” models may amplify bias, misinformation, and security flaws.

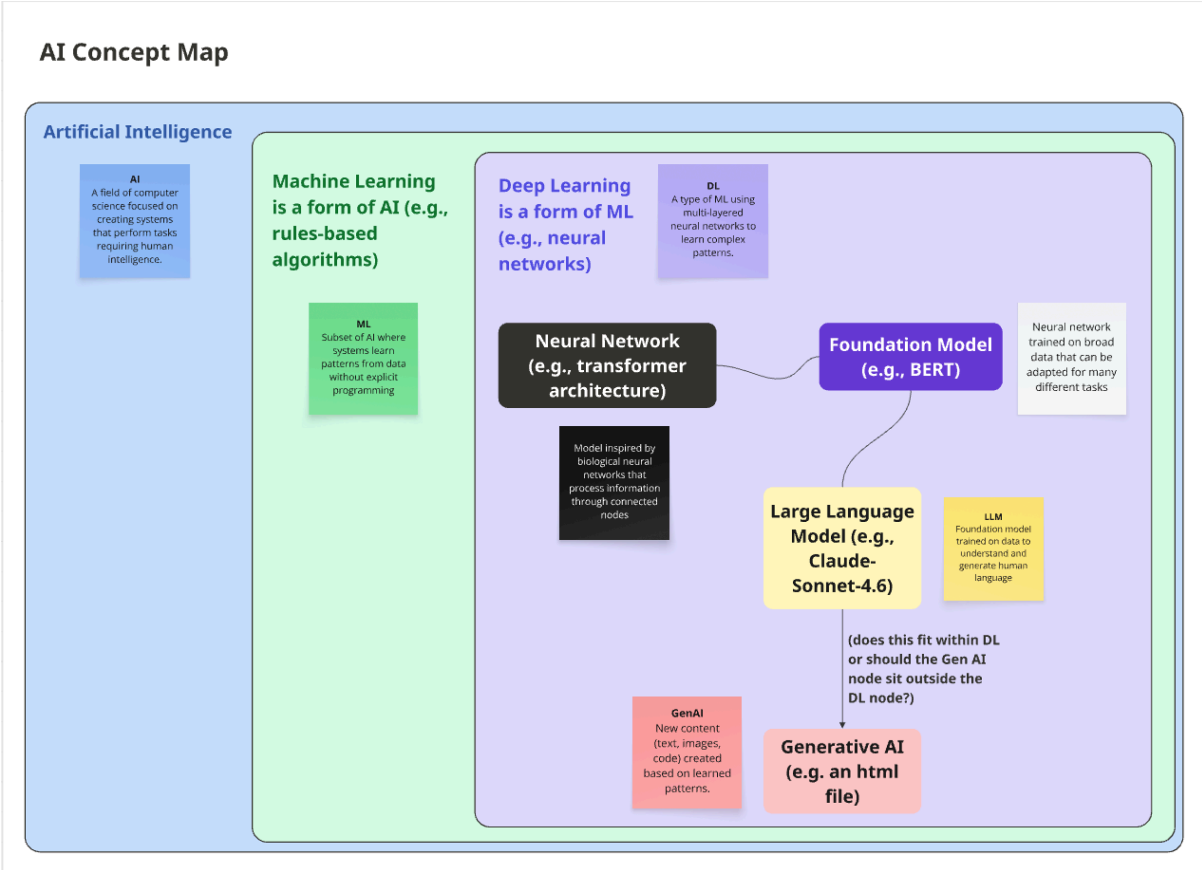
		flaws can be amplified.	
Bommasani et al. Section 1, page 7	Foundation models don't exist without people.	People are the source of data for training the models, and people are the recipients of output generated by the models.	We should be careful about what we put into the model and critical about what the model produces as output.
Bommasani et al. Section 2, page 9	Models are trained to perform a task, but there are rules.	Models may be capable of creating toxic content. Should they still be made available to the public?	This makes me think about Anthropic slowing the release of Mythos and Trump requiring his review before models are released to the public.
Bommasani et al. Section 2, page 10	AI frontier models are being developed by industry, where there is little incentive to help the greater good.	Analogy to pharmaceutical companies not investing in developing cure for malaria.	Universities are better suited to explore how AI can provide large social benefit.
Bommasani et al. Section 2, pages 14-20	This report was submitted to/published in arXiv in July 2022 and so many of the points raised remain valid in June 2026 (including the debate about data centers!).	Multiple capabilities reviewed in Overview. Transformer architecture was the biggest change. Everything is impacted by deployment of the resulting models.	The potential for model use is unlimited. As an academic, I want to use the models responsibly (for the betterment of society).

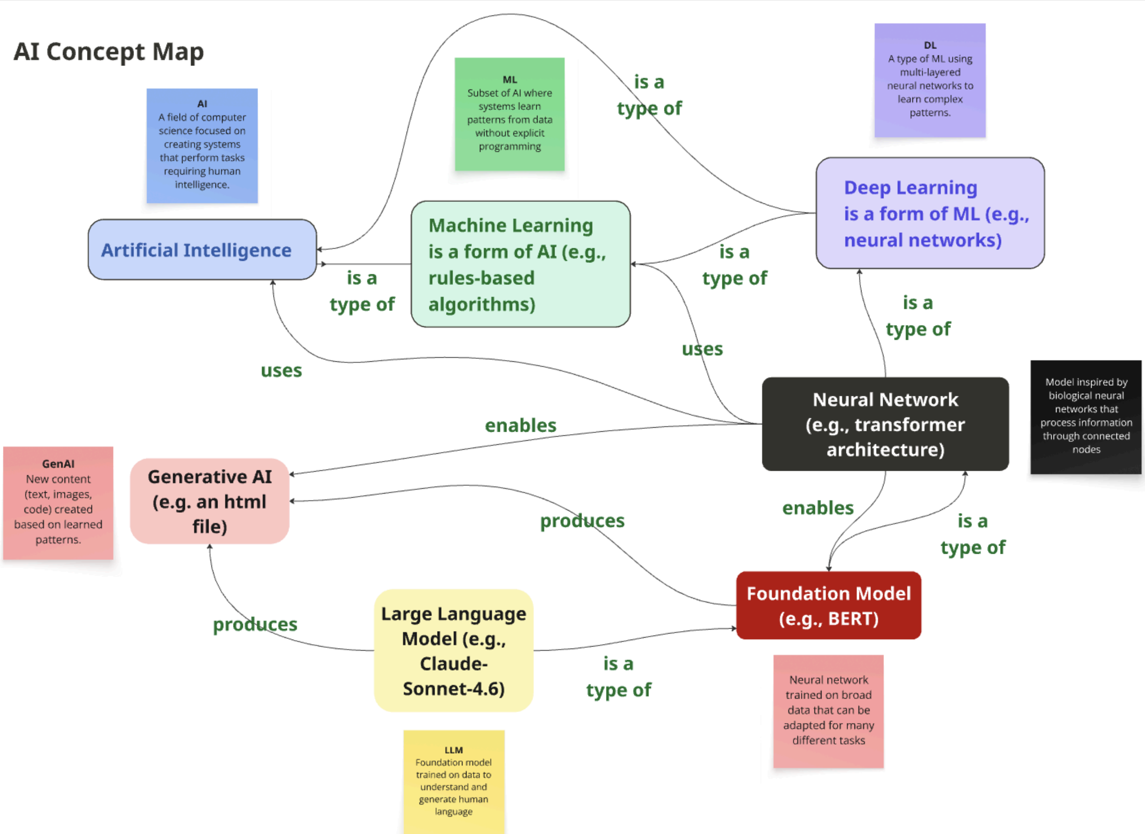
AI Index Executive Summary	The US is putting all of it’s R&D into AI, at the expense of other areas for research, education, social net.	Top takeaway #6	Elections make a difference.
AI Index Chapter 4, page 173, 192	More people are using GenAI, but they are primarily using free tools. Financial pressures on the frontier model companies may not allow free usage (although they get our data as payment) for much longer.	Value to consumers is growing, although I don’t fully understand how that’s calculated. There is some explanation on page 192 (how much compensation they would accept to give up access to all generative AI tools for one month). The report states, “the average consumer surplus among U.S. generative AI users increased by 27% from $98 in 2025 to $125 by March 2026, while the median value per user tripled, from $3.40 to $11.40 over the same period.”	Be prepared to pay for AI.

AI Index Chapter 6, page 255, 272	On clinical reasoning tasks, leading AI models now score higher than most physicians on structured clinical evaluations, yet nearly half of clinical AI studies still rely on simulated scenarios rather than real patient data. Autonomous or semiautonomous agents are a new framework for clinical applications.	"Multiagent frameworks, in which multiple AI agents take on specialized roles—such as diagnostician or pharmacist—and collaborate through structured reasoning protocols, have shown early promise on benchmark evaluations."	There is a lot of interest in deploying AI and agents for diagnostic purposes. Benchmarks are being created and studies are being done to assess accuracy in real clinical scenarios. How will this affect the future role of clinicians?

Notes from Task 2: Personal Concept Map Construction [45 min]

Made with Miro.com


AI Concept Map
Artificial Intelligence
AI
A field of computer science focused on creating systems that perform tasks requiring human intelligence.
Machine Learning is a form of AI (e.g., rules-based algorithms)
ML
Subset of AI where systems learn patterns from data without explicit programming
Deep Learning is a form of ML (e.g., neural networks)
DL
A type of ML using multi-layered neural networks to learn complex patterns.
Neural Network (e.g., transformer architecture)
Model inspired by biological neural networks that process information through connected nodes
Foundation Model (e.g., BERT)
Neural network trained on broad data that can be adapted for many different tasks
Large Language Model (e.g., Claude-Sonnet-4.6)
LLM
Foundation model trained on data to understand and generate human language
(does this fit within DL or should the Gen AI node sit outside the DL node?)
GenAI
New content (text, images, code) created based on learned patterns.
Generative AI (e.g. an html file)


AI Concept Map
AI
A field of computer science focused on creating systems that perform tasks requiring human intelligence.
Artificial Intelligence
ML
Subset of AI where systems learn patterns from data without explicit programming
Machine Learning is a form of AI (e.g., rules-based algorithms)
is a type of
is a type of
DL
A type of ML using multi-layered neural networks to learn complex patterns.
Deep Learning is a form of ML (e.g., neural networks)
is a type of
is a type of
uses
uses
Neural Network (e.g., transformer architecture)
Model inspired by biological neural networks that process information through connected nodes
enables
GenAI
New content (text, images, code) created based on learned patterns.
Generative AI (e.g. an html file)
produces
enables
is a type of
Foundation Model (e.g., BERT)
Neural network trained on broad data that can be adapted for many different tasks
produces
Large Language Model (e.g., Claude-Sonnet-4.6)
is a type of
LLM
Foundation model trained on data to understand and generate human language

Notes from Task 3: Structured LLM Exploration — Five-Prompt Protocol [45 min]

Step-by-step instructions:

1. Open Claude.ai or Google Gemini free tier. Start a brand new conversation — do not use an existing one.
2. Run each of the five prompts below VERBATIM. Do not modify the wording — exact replication enables cohort comparison.
3. After each prompt, immediately complete one row in the Exploration Log before moving to the next.
4. PROMPT 1 — Knowledge cutoff: 'What is the current prime minister of Canada? Please cite the source of your information and state your confidence level.'
5. PROMPT 2 — Summarization: 'Summarize the following passage in exactly three bullet points, one sentence each: [PASTE: the opening paragraph of Bommasani et al. Section 1 — the paragraph beginning with the word We]'
6. PROMPT 3 — Arithmetic reasoning: 'Calculate the following and show every step: (17 × 23) + (141 ÷ 3). Then verify your answer by computing it a second way.'
7. PROMPT 4 — Explanation quality: 'Explain what a large language model is to a curious 12-year-old who has never heard the term. Use one analogy.'
8. PROMPT 5 — Epistemic limits: 'What are the three most significant AI developments that occurred in the last 14 days? Cite your sources.'
9. Optional: If time allows, repeat the protocol with a second LLM tool and note differences in your Observation column.

Five-Prompt Exploration Log:

Prompt #	Tool used	Output summary (less than 40 words)	Accuracy (1-5)	Usefulness (1-5)	Key observation
1	Claude Sonnet	Mark Carney; official Canadian	5	5	This was straightforward and correct

	4.6 6/19/26	government website (pm.gc.ca), Wikipedia's article on the Prime Minister of Canada, and the Globe and Mail			
2	Claude Sonnet 4.6 6/19/26	Three sentences about the content, just as I requested.	5	4	The summary seems good to me, although I'm still learning the termininoloyg and nuance.
3	Claude Sonnet 4.6 6/19/26	438; calculation was indeed shown step by step in 2 different ways	5	5	I arrived at the same answers independently.
4	Claude Sonnet 4.6 6/19/26	A reasonable attempt at the explanation and analogy.	3	2	I can't imagine anyone reading everything, so it wasn't a great start, but the concept may be the way humans acquire language. The autocomplete analogy is one I've heard often, so it's not bad.

5	Claude Sonnet 4.6 6/19/26	6/9/26 release and ban of Claude Fable 5; OpenAI acquisition of Astral; Northwestern Brain-AI Breakthrough & China's $295B Infrastructure Plan	4	4	It actually gave me 4, since #3 included 2 developments. This is only moderately useful since I don't know if the answer is correct without being personally current on the news. There could be bias in the reporting. The model seems to be drawing from a small pool of sources. I found information from multiple sources stating that OpenAI announced it will acquire Astral on March 19, 2026, so it wasn't in the last 14 days, although who knows when the actual contract was signed.

Notes from Task 4: Reflection Journal — Entry 1 [30 min]

Step-by-step instructions:

1. Open your journal document. Write a header: 'Module 1 Reflection | [Your Name] | [Date]'
2. Write continuously for 25 minutes. Address each of the three prompts below in sequence.
3. Aim for 80–130 words per prompt (250–400 words total). Use the sentence starters if you are stuck.
4. In the final 5 minutes, re-read and underline one sentence you want to remember or discuss.
5. Do not edit for style. This is a thinking document — clarity of idea matters; polish does not.

Reflection prompts — respond to all three:

PROMPT A:
What assumption about AI did this module confirm, challenge, or complicate? Describe the assumption and what changed. (Sentence starter: 'Before this module, I assumed AI... After reading/exploring, I now understand...')
Response: Before this module, I assumed AI was rooted in academic research and it blew up once it was realized there was money to be made from it. I heard from academic colleagues that it became hard to get funding for academic research on AI since the big companies could throw more money, people, and time at any interesting problem or promising solution. After reading/exploring, I now understand there may be an important role for academic institutions in framing how AI is applied. I am interested in how AI can be used to improve student learning. I'm frustrated by the big companies giving away AI tools to students for free, but not making the same offer to faculty. Faculty need time and access to explore what AI can do so it can enhance learning, rather than preventing learning.

PROMPT B:
What one aspect of how LLMs work technically surprised you most — and why does it matter for your professional context? (Sentence starter: 'I was surprised to learn that LLMs... This matters for my work because...')
Response: I was surprised to learn that LLMs can be effectively trained by self-supervised learning (using unannotated data). I understand the pretrained

(labeled/annotated by humans) models, but realize this is a time-consuming task. I feel less comfortable with self-supervised learning because it seems more likely to contain misinformation. I realize people can mislabel items, so it’s not a guarantee that one method is better than the other. Proper labeling matters for my work because LLMs in fields like radiology and pathology are highly dependent on quality data. Errors, whether introduced by humans or algorithms, reduce the confidence in the output

PROMPT C:
Name one specific recurring task in your role that you now think could be affected by current AI. Be concrete: name the task, identify the AI mechanism, and describe one realistic impact scenario — positive or cautionary. (Sentence starter: 'One task I perform regularly is [specific task]. Based on what I learned, [specific AI mechanism] could [specific impact]...')
Response: One task I perform regularly is grading assignments. Based on what I learned, AI could help me organize and evaluate student submissions. I’m not very good yet at creating yes/no rubrics, but I think they could be useful for my purposes. I’m still wary of allowing AI to provide individual feedback to students, but I find it useful for providing group feedback. Students are still learning how to use AI effectively, so there is a lot of variation between students. As an instructor, I want to develop strategies to help students learn skills that can improve their learning experience.